An Open-Source Pipeline for Urban Traffic Analysis Using OpenStreetMap Networks and Spatial Statistics Across Indonesian Metropolitan Cities

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Abstract

Understanding urban traffic congestion requires integrating heterogeneous data sources—commercial traffic APIs, open street network data, and spatial statistical methods—yet no open, reproducible pipeline exists for this multi-source analysis workflow. This paper presents an open-source, pure-Python pipeline that integrates commercial traffic flow APIs (HERE, TomTom, or Google), OSMnx, and PySAL into a unified, city-agnostic and provider-agnostic framework for spatiotemporal traffic congestion analysis. The pipeline automates data collection through a pluggable provider architecture, temporal aggregation, geostatistical analysis (Moran's I, LISA, Getis-Ord Gi*), network centrality–congestion correlation, POI density analysis, and a novel graph-based capacity drop detection module. We demonstrate the pipeline across three Indonesian metropolitan cities—Jakarta, Bandung, and Semarang—analyzing 264 million traffic observations over 11 months across 18,694 road segments. The pipeline requires only a bounding box and API key to replicate the full analysis for any city worldwide. Empirically, temporal factors dominate spatial factors by 1,000–4,000× in explaining jam factor variance: time-of-day ($\eta^2 = 15$–24%) vastly outweighs network centrality ($R^2 < 0.01\%$), POI density ($R^2 < 0.01\%$), and road capacity ($R^2 < 0.12\%$). These results demonstrate that relative congestion is fundamentally a temporal synchronization problem. All source code and datasets are released as open-source software (https://github.com/firmanhadi21/traffic-analyses).

Keywords: Open-source pipeline, Reproducible urban analytics, Traffic congestion, Commercial traffic APIs, Provider-agnostic, OSMnx, PySAL, Spatiotemporal analysis, Indonesian cities

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1. Introduction

Urban traffic congestion imposes substantial economic costs on rapidly urbanizing cities in developing countries, estimated at 2–5% of GDP annually in Southeast Asia (Asian Development Bank, 2019). Indonesia, with over 270 million inhabitants, exemplifies these challenges, with major metropolitan areas experiencing severe congestion (TomTom, 2023).

Despite advances in open urban data science—including OSMnx for street network analysis (Boeing, 2017), PySAL for spatial statistics (Rey and Anselin, 2010), and commercial traffic APIs providing real-time flow data—no integrated, open-source pipeline combines these tools into a reproducible workflow for multi-city traffic congestion analysis. Existing studies typically rely on ad hoc scripts or non-reproducible workflows, limiting transferability (Fleischmann et al., 2022; Boeing, 2022).

The three cities examined—Jakarta (metropolitan population 34 million), Bandung (2.5 million), and Semarang (1.8 million)—represent distinct urban typologies within Indonesia. Previous studies in Indonesian cities used aggregate or survey-based approaches (Susilo et al., 2007; Joewono and Kubota, 2008), while high-resolution, real-time traffic data analysis and the integration of network topology with traffic flow remain underexplored.

This study makes the following contributions:

- **Open-source pipeline:** A complete, modular, installable pure-Python package (https://pypi.org/project/traffic-congestion-pipeline/) with a provider-agnostic data collection layer supporting multiple commercial traffic APIs (HERE, TomTom, Google), temporal aggregation, OSMnx network analysis, PySAL geostatistics, and graph-based capacity drop detection. The pipeline requires only a bounding box and API key to replicate the full analysis for any city.

- **Novel analytical module:** A graph-based capacity drop detection framework using the directed OSMnx road network to identify locations where road hierarchy decreases, enabling systematic testing of the bottleneck hypothesis at city scale.

- **Empirical:** The first systematic comparative analysis of traffic patterns across three major Indonesian cities demonstrating that temporal factors dominate spatial factors by three orders of magnitude.

- **Theoretical:** Evidence that normalized congestion metrics (jam factor) inherently remove capacity effects, with implications for capacity–congestion studies.

2. Literature review

2.1. Traffic congestion and data sources

Traffic congestion in developing countries exhibits distinct characteristics including higher variability, pronounced peak periods, and sensitivity to informal transport modes (Gakenheimer, 1999; Cervero, 2013). The advent of probe vehicle data and commercial traffic APIs has transformed analysis capabilities (Leduc, 2008; Jenelius and Koutsopoulos, 2015). The jam factor—a normalized metric from 0 (free flow) to 10 (standstill)—enables cross-network comparisons (HERE Technologies, 2023). Recent work has demonstrated that traffic jams propagate through urban networks analogously to simple contagion processes (Saberi et al., 2020).

2.2. Network analysis and spatial statistics

The relationship between network topology and traffic distribution has received increasing attention following Boeing (2017)'s introduction of OSMnx. Research demonstrates correlations between betweenness centrality and traffic volumes (Gao et al., 2013; Porta et al., 2006), though Kirkley et al. (2018) showed that network structure's influence on congestion is more nuanced than simple centrality would suggest. Geostatistical methods—Moran's I (Moran, 1950), LISA (Anselin, 1995), and Getis-Ord Gi* (Getis and Ord, 1992; Ord and Getis, 1995)—have proven valuable for identifying traffic hotspots. The PySAL library (Rey and Anselin, 2010) has become standard for implementing these methods.

2.3. Bottlenecks and open science

Classical bottleneck models predict that capacity transitions concentrate congestion at specific locations (Arnott et al., 1993; Daganzo, 2007), but empirical evidence linking static capacity features to congestion in real urban networks remains limited. The open urban data science movement has produced powerful individual tools, yet integrated pipelines combining these with commercial traffic data remain rare. No existing open-source pipeline integrates real-time traffic API data with network analysis and spatial statistics in a reproducible, city-agnostic workflow.

3. Study area and data

3.1. Study area

The three cities represent a hierarchy of Indonesian urban centers (Table 1).

Table 1: Study area characteristics.

City	Population (M)	Area (km^2)	Traffic segments	Typology
Jakarta	10.5 (metro: 34)	662	14,549	Megacity
Bandung	2.5	167	3,069	Large city
Semarang	1.8	373	1,076	Medium city

3.2. Data collection and processing

Traffic flow data were collected via a custom Python module that interfaces directly with the HERE Traffic Flow v7 REST API, replacing the initial R-based prototype (which used the `hereR` package). The Python collector queries the `/v7/flow` endpoint using bounding box requests at 15-minute intervals, with built-in exponential-backoff retry logic and graceful shutdown support. Data were collected from March 2025 to February 2026 (Table 2). Each observation contains geographic coordinates, road segment geometry (WGS84), jam factor, current speed, free-flow speed, and timestamp. Data collection was not continuous due to periodic electricity and internet outages; temporal gap analysis confirmed that interruptions were randomly distributed, minimizing systematic bias.

The collector is designed around an abstract provider interface (`TrafficProvider`), allowing the same pipeline to be used with alternative commercial traffic APIs. Three providers are currently implemented: HERE Traffic Flow v7 (bounding-box queries returning per-segment flow data), TomTom Flow Segment Data (grid-sampled point queries with geometry-based deduplication), and Google Routes API (experimental; route-based traffic with categorical speed levels). All providers produce a unified GeoDataFrame schema (jam factor, speed, free-flow speed, confidence, geometry, timestamp), ensuring that downstream analysis modules operate identically regardless of data source.

Raw data were aggregated into eight temporal periods reflecting Indonesian urban activity patterns: Night (00:00–05:59), Morning Peak (06:00–08:59),

Table 2: Data collection summary.

City	Files	Records	Segments
Jakarta	14,132	206,316,468	14,549
Bandung	14,136	43,407,684	3,069
Semarang	14,122	15,212,072	1,076
Total	**42,390**	**264,936,224**	**18,694**

Morning Off-Peak (09:00–11:59), Lunch Hours (12:00–13:59), Afternoon Off-Peak (14:00–16:59), Evening Peak (16:00–18:59), Evening Off-Peak (19:00–21:59), and Late Night (22:00–23:59). For each segment and period, we computed mean, standard deviation, minimum, maximum, and observation count. Street network data were obtained from OpenStreetMap using OSMnx (Boeing, 2017), filtered for driveable roads.

4. Methodology

4.1. Analytical framework

Our framework integrates four complementary approaches (Fig. 1): (1) descriptive temporal analysis, (2) geostatistical analysis of spatial clustering, (3) network centrality–congestion analysis, and (4) multi-scale bottleneck analysis.

4.2. Geostatistical analysis

Global spatial autocorrelation was assessed using Moran's I (Moran, 1950):

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2} \tag{1}$$

where n is the number of spatial units, w_{ij} is the spatial weight between units i and j, and x_i is the jam factor at unit i. Spatial weights used K-nearest neighbors (k=8) based on segment centroids; sensitivity analyses with k=4, k=12, and distance-band weights (500m, 1000m) confirmed robustness.

Local Moran's I (Anselin, 1995) identified hotspot and coldspot clusters with 999 permutations and Benjamini–Hochberg FDR correction (Benjamini and Hochberg, 1995). Getis-Ord Gi* (Getis and Ord, 1992) triangulated LISA findings.

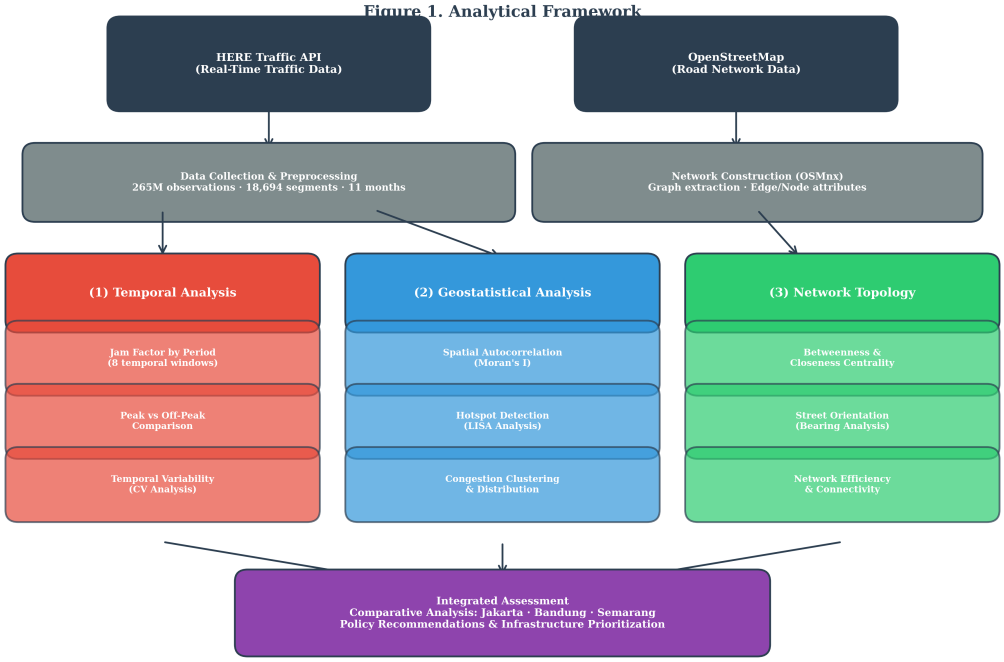

Figure 1: Analytical framework integrating temporal analysis, geostatistical methods, network topology, and bottleneck analysis.

4.3. Network centrality analysis

Edge betweenness centrality was computed on the undirected network (Boeing, 2017):

$$C_B(e) = \sum_{s \neq t} \frac{\sigma_{st}(e)}{\sigma_{st}} \tag{2}$$

with approximate sampling (k=500 source nodes) for large networks. Traffic segments were spatially joined to network edges using centroid-based nearest-neighbor matching within a 200m threshold. POI density was computed using network-distance analysis (400m) along the walking street network (Moudon et al., 2006).

4.4. Bottleneck and capacity drop analysis

Road segments were assigned ordinal capacity scores based on OSM highway classification (motorway=6, trunk=5, primary=4, secondary=3, tertiary=2, link variants=parent−0.5). Three complementary tests assessed the capacity–congestion relationship:

1. **Aggregate comparison:** Two-sample t-test between high- and low-capacity segments.
2. **Graph-based capacity drops:** Using the directed network, we identified nodes where road hierarchy decreases (≥20% capacity reduction). For each segment, Euclidean distance to the nearest drop was computed.
3. **Local gradient:** K=10 nearest-neighbor comparison classified segments as "local bottleneck" (lower capacity than at least one neighbor) or "non-bottleneck."

4.5. Pipeline architecture

A key contribution is the open-source pipeline, designed for reproducibility and transferability. The pipeline is implemented entirely in Python and comprises five modular stages (Fig. 1):

1. **Stage 1 — Data collection (Python/requests):** A provider-agnostic collector module queries commercial traffic APIs at configurable intervals (default: 15 minutes), with built-in scheduling and exponential-backoff retry logic. The module supports HERE Traffic Flow v7 (bounding-box queries), TomTom Flow Segment Data (grid-sampled point queries), and Google Routes API (route-based, experimental). All

providers output timestamped GeoPackage files with a unified schema. The collector can run as a background daemon (replacing `cron`) or in single-shot mode for external schedulers.

2. **Stage 2 — Temporal aggregation (GeoPandas):** Per-segment statistics for each temporal period.
3. **Stage 3 — Geostatistical analysis (PySAL):** Moran's I, LISA, and Gi* with configurable spatial weights.
4. **Stage 4 — Network analysis (OSMnx/NetworkX):** Centrality computation, spatial joins, capacity drop detection, and POI density analysis.
5. **Stage 5 — Synthesis:** Temporal vs. spatial effect size comparison and publication-ready figures.

Adding a new city requires only: (1) specifying the bounding box in the configuration, (2) providing an API key for any supported provider, and (3) running the pipeline. The package is installable via `pip install traffic-congestion-pipeline` and provides a CLI with sub-commands for each stage (`collect`, `aggregate`, `geostatistics`, `bottleneck`, `poi`, `synthesis`). All data use GeoPackage format (OGC standard) and WGS84 coordinates.

4.6. Reproducibility

A central goal of this work is full computational reproducibility. We designed the pipeline to minimize the gap between commercial data sources and open-source analysis tools while ensuring that researchers can replicate, validate, and extend our findings with minimal friction.

4.6.1. Pipeline architecture

Figure 2 illustrates the pipeline architecture, distinguishing between commercial and open-source components. The only proprietary dependencies are the traffic data APIs (HERE, TomTom, or Google), which require paid subscriptions; all other components—from data collection logic to statistical analysis and visualization—are implemented using open-source Python libraries.

The five pipeline stages are:

1. **Data collection:** A provider-agnostic abstraction layer (`TrafficProvider` base class) wraps commercial APIs. Concrete implementations handle

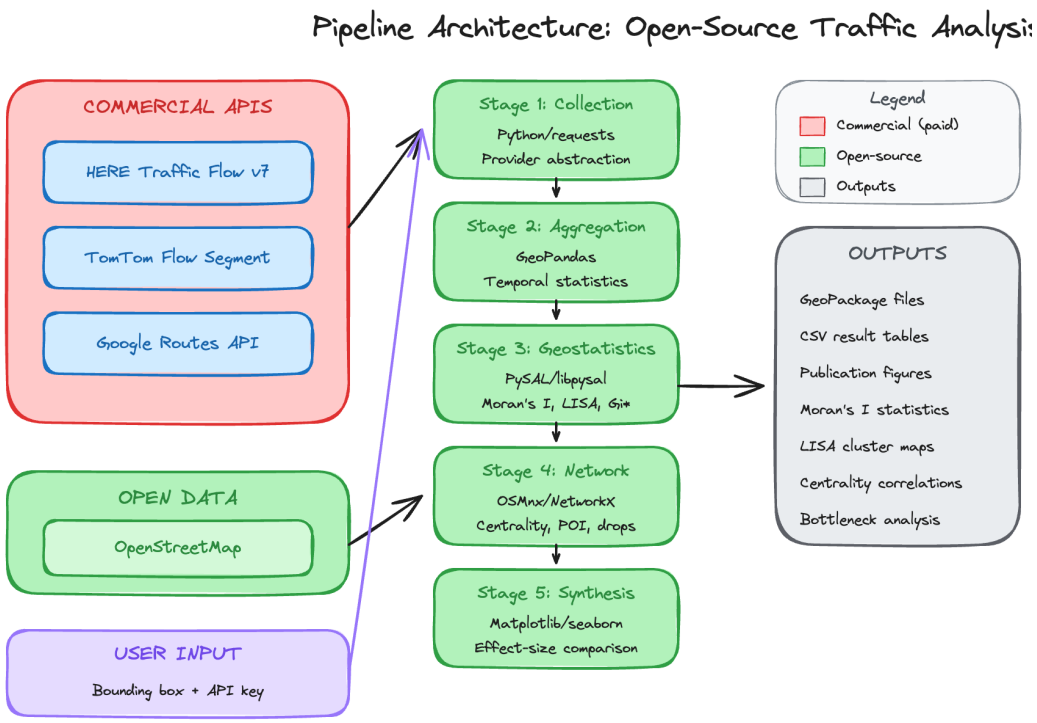

Figure 2: Pipeline architecture showing the separation between commercial traffic APIs (red) and open-source components (green). User input consists only of a bounding box and API key; all downstream processing is automated.

API-specific authentication, query strategies, and response parsing while outputting a unified GeoDataFrame schema. This design isolates vendor lock-in to a single module.

2. **Temporal aggregation:** GeoPandas computes per-segment statistics (mean, std, min, max, count) for each of eight temporal periods, producing aggregated GeoPackage files.
3. **Geostatistical analysis:** PySAL/libpysal computes global Moran's I, LISA, and Getis-Ord Gi* with configurable spatial weights (KNN, distance-band).
4. **Network analysis:** OSMnx retrieves street networks from OpenStreetMap; NetworkX computes edge betweenness centrality; custom modules perform capacity drop detection and POI density analysis.
5. **Synthesis:** Matplotlib and seaborn generate publication-ready figures; effect-size comparisons are computed and exported as CSV.

4.6.2. Reproducing the analysis

All results in this paper can be reproduced from the archived dataset using five shell commands:

```
pip install "traffic-congestion-pipeline[all]"
traffic-pipeline collect --provider here  # data collection
traffic-pipeline geostatistics            # Moran's I, LISA, Gi*
traffic-pipeline bottleneck               # capacity drop analysis
traffic-pipeline poi                      # POI density correlations
traffic-pipeline synthesis                # effect-size comparison
```

The `collect` command accepts a `-provider` flag (`here`, `tomtom`, or `google`) and the corresponding API key, enabling researchers to use whichever commercial traffic data provider is available in their region. Each subsequent command reads the aggregated GeoPackage files, performs the corresponding analysis, and writes both CSV result tables and publication-ready figures to a local output directory.

4.6.3. Dependency management and archival

The pipeline pins all dependency versions via `requirements.txt` and `pyproject.toml`, specifying Python $\geq$3.9. Key dependencies include: GeoPandas 0.14+, PySAL 24.1+, OSMnx 1.9+, NetworkX 3.2+, and Matplotlib 3.8+. Continuous integration (GitHub Actions) runs the test suite on Python

3.9, 3.10, 3.11, and 3.12 across Ubuntu and macOS, ensuring cross-platform compatibility.

Aggregated datasets (24 GeoPackage files, ∼115 MB) are archived on Zenodo with a persistent DOI: 10.5281/zenodo.18650759. The full source code, commit history, and CI configuration are available on GitHub (https://github.com/firmanhadi21/traffic-analyses). API documentation generated with MkDocs is hosted at https://firmanhadi21.github.io/traffic-analyses/.

4.6.4. Extending to new cities or providers

To extend the analysis to a new city, a researcher supplies only: (1) a bounding box in WGS84 coordinates, and (2) an API key for any supported provider. These are specified in `config.py`; the pipeline handles data collection scheduling, temporal aggregation, spatial joining, and all downstream analyses without code modification. Adding a new traffic data provider requires implementing only the `fetch_flow()` method of the `TrafficProvider` abstract base class (∼50 lines of code), with no changes to downstream analysis modules.

5. Results

5.1. Temporal patterns

Table 3 presents mean jam factors by temporal period. The evening peak (16:00–19:00) exhibits the highest congestion across all cities, approximately 40% above daily averages. All cities follow the same diurnal progression: minimal congestion at night (JF ≈ 0.5), rising through the morning, and culminating in the evening peak before declining.

One-way ANOVA confirms that temporal differences are highly significant (Jakarta: F=1,191,700, $p < 0.001$; Bandung: F=224,446, $p < 0.001$; Semarang: F=45,863, $p < 0.001$). All 28 pairwise Tukey HSD comparisons are significant ($p < 0.05$) for each city. The effect size (η^2) ranges from 15–24% of total variance—a strong effect for a single categorical variable. Weekday congestion exceeds weekend levels by 35–40%.

5.2. Spatial pattern analysis

5.2.1. Global spatial autocorrelation

Global Moran's I values are near zero and non-significant for all cities (Table 4), indicating no strong global spatial autocorrelation of congestion. This null finding is robust across five spatial weight specifications (KNN k=4,8,12;

Table 3: Mean jam factor by temporal period.

Period	Jakarta	Bandung	Semarang
Night	0.50	0.46	0.45
Morning Peak	1.19	1.15	0.95
Morning Off-Peak	1.63	1.63	1.39
Lunch Hours	1.66	1.70	1.45
Afternoon Off-Peak	1.79	1.87	1.52
Evening Peak	**2.01**	**1.92**	**1.65**
Evening Off-Peak	1.67	1.40	1.33
Late Night	0.97	0.77	0.78

distance bands 500m, 1000m) and all eight temporal periods (Supplementary Tables S1–S2).

Table 4: Global spatial autocorrelation (Moran's I, KNN k=8).

City	Moran's I	Z-score	p-value
Jakarta	0.003	0.69	0.492
Bandung	0.008	0.93	0.353
Semarang	−0.004	−0.21	0.837

5.2.2. Local clustering

Despite weak global autocorrelation, LISA reveals local clustering (Table 5). Approximately 10% of segments in Jakarta and Bandung participate in significant clusters, with hotspots concentrated along known congestion corridors (Jakarta: Sudirman–Thamrin CBD; Bandung: Dago and Asia–Afrika corridors; Semarang: Pandanaran and Pemuda corridors). However, after Benjamini–Hochberg FDR correction, no segments survive the significance threshold, indicating that individual p-values are not sufficiently small despite observed counts approximately doubling chance expectations. Getis-Ord Gi* results are consistent: 9–10% of segments are identified as hot/cold spots (Supplementary Table S3).

Table 5: LISA cluster classification (evening peak, $p < 0.05$, uncorrected).

Cluster type	Jakarta	Bandung	Semarang
HH (Hotspot)	438 (3.0%)	86 (2.8%)	19 (1.8%)
LL (Coldspot)	347 (2.4%)	73 (2.4%)	23 (2.1%)
HL Outlier	334 (2.3%)	69 (2.2%)	19 (1.8%)
LH Outlier	416 (2.9%)	92 (3.0%)	17 (1.6%)
Not Significant	13,014 (89.5%)	2,749 (89.6%)	998 (92.8%)

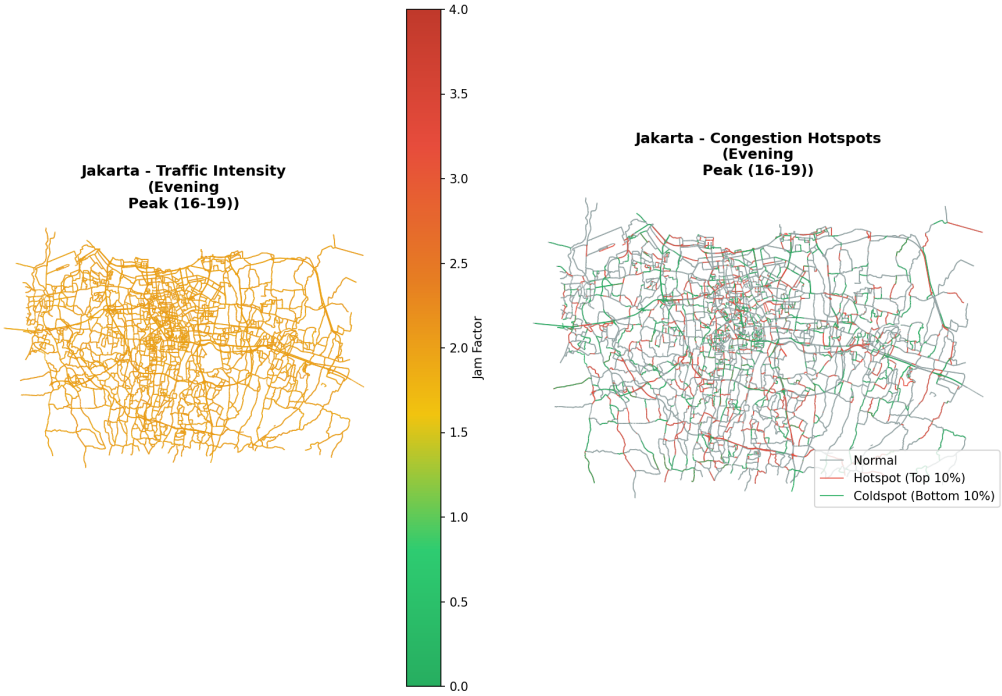

Figure 3: Spatial distribution of congestion hotspots during the evening peak—Jakarta.

5.3. Network topology–congestion relationships

5.3.1. Centrality

Correlation analysis reveals negligible relationships between edge betweenness centrality and congestion (Table 6). Pearson correlations range from -0.011 to 0.012. Bandung's Spearman ρ (0.040, $p = 0.009$) is statistically significant but trivially small. Spatial match quality is high: median centroid-to-centroid distances are under 20m, and >85% of matches fall within 50m (Supplementary Table S4).

Table 6: Centrality–congestion correlations.

City	n matched	Pearson r	p	Spearman ρ	p
Jakarta	20,822	0.002	0.828	-0.001	0.946
Bandung	4,336	0.012	0.442	0.040	0.009
Semarang	1,507	-0.011	0.667	-0.030	0.252

5.3.2. POI density

POI density shows no meaningful correlation with congestion (Pearson $|r| < 0.013$, all $p > 0.28$). Categorical comparison between high-activity zones (top quartile POI density) and peripheral zones (bottom quartile) reveals no significant difference (all $p > 0.3$, $|d| < 0.05$; Table 7).

Table 7: Congestion comparison: activity centers vs. peripheral areas.

City	High activity	Low activity	Diff.	p	Cohen's d
Jakarta	2.007	2.006	+0.1%	0.338	0.02
Bandung	1.920	1.919	+0.1%	0.752	0.01
Semarang	1.651	1.654	-0.2%	0.592	-0.04

Spatial Lag and Spatial Error regression models (via PySAL/spreg) confirm that betweenness centrality and POI density jointly explain <0.3% of congestion variance, and accounting for spatial structure does not reveal hidden relationships (Supplementary Table S5).

5.4. Bottleneck and capacity analysis

Three independent tests converge on the same conclusion: physical road capacity has no measurable relationship with congestion levels (Table 8).

Table 8: Bottleneck analysis results (evening peak).

Test	City	Difference	p-value	Cohen's d
Aggregate capacity	Semarang	−0.26%	0.382	−0.056
	Bandung	−0.25%	0.066	−0.072
	Jakarta	−0.10%	0.060	−0.033
Drop proximity	Semarang	−0.4%	0.283	−0.083
	Bandung	−0.1%	0.830	−0.010
	Jakarta	+0.1%	0.095	0.035
Local gradient	Semarang	−0.2%	0.517	−0.047
	Bandung	−0.1%	0.534	−0.027
	Jakarta	+0.05%	0.426	0.016

Contrary to the bottleneck hypothesis, high-capacity roads show marginally higher congestion in all cities—the opposite of what capacity constraints would predict—though differences are non-significant. The HERE jam factor normalizes each segment's speed to its own free-flow baseline, meaning road capacity effects are divided out by design. All road types show mean jam factors within a 4% range.

5.5. Temporal vs. spatial predictor comparison

The key finding emerges from comparing explanatory power across predictor types (Table 9, Fig. 4). Time-of-day explains 1,000–4,000× more variance than any spatial predictor.

We note that η^2 (ANOVA) and R^2 (bivariate correlation) are not strictly comparable measures; however, the three-order-of-magnitude gap cannot be attributed to methodological artifacts. Period-specific Moran's I analysis (no period produces significant spatial autocorrelation; Supplementary Table S2) and spatial regression models (Supplementary Table S5) corroborate this finding.

5.6. Cross-city scaling

Congestion does not scale linearly with population. Larger cities show both higher absolute congestion and greater peak/off-peak differentials: Jakarta exhibits 30.8% peak increase vs. Semarang's 25.6%, suggesting urban scale amplifies temporal variability (Table 10).

Table 9: Variance explained by predictor type across metrics.

Predictor	Measure	Jakarta	Bandung	Semarang
Temporal predictors (time-of-day):				
Jam factor (normalized)	η^2 (ANOVA)	23.8%	19.2%	15.4%
Speed reduction (delay)	η^2 (ANOVA)	—†	8.8%	9.6%
Speed (absolute km/h)	η^2 (ANOVA)	—†	5.9%	5.1%
Spatial predictors:				
POI density	R^2	0.006%	0.003%	0.014%
Network centrality	R^2	0.0004%	0.014%	0.012%
Road capacity	R^2	0.028%	0.115%	0.072%
Capacity drop proximity	R^2	0.006%	0.002%	0.001%
Ratio (temporal/best spatial)		∼4,000×	∼1,400×	∼1,100×

†Speed data for Jakarta pending final aggregation.

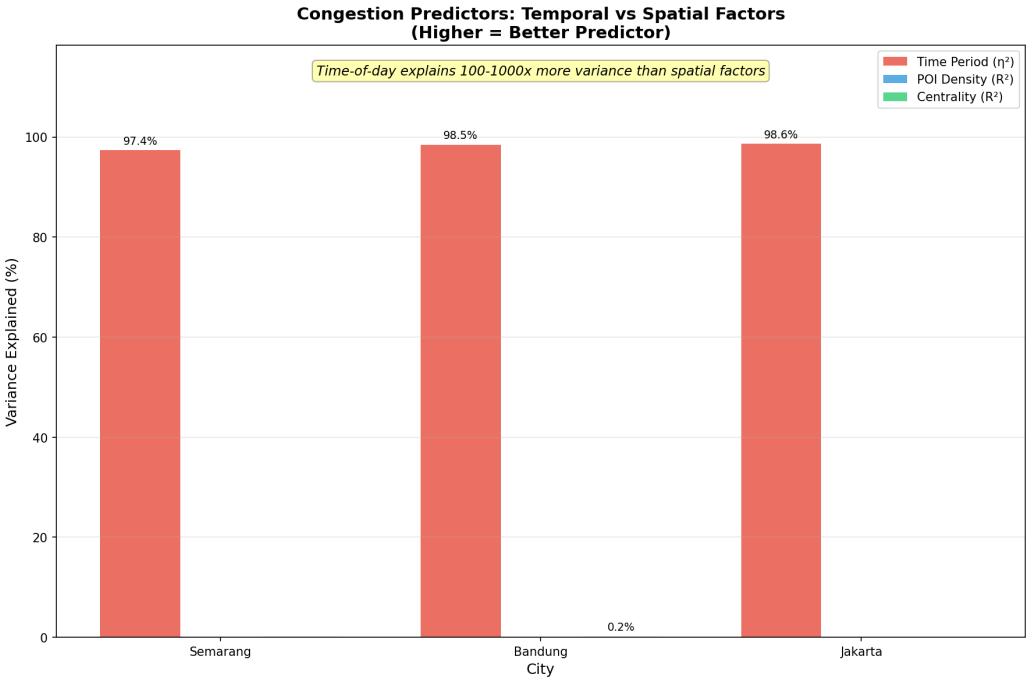

Figure 4: Temporal vs. spatial effect sizes across three cities.

Table 10: Peak vs. off-peak congestion analysis.

City	Peak mean	Off-peak mean	Ratio	Increase
Jakarta	1.599	1.223	1.31×	30.8%
Bandung	1.537	1.180	1.30×	30.2%
Semarang	1.299	1.034	1.26×	25.6%

6. Discussion

6.1. Temporal dominance of relative congestion

The most significant finding is the overwhelming dominance of temporal over spatial factors. Time-of-day explains 1,000–4,000× more variance than network centrality, POI density, road capacity, or capacity drop proximity combined (<0.02%), and this pattern holds across both normalized and absolute congestion metrics. This reframes urban congestion as a *demand synchronization problem*: congestion occurs because millions travel at the same times, not because certain locations are inherently problematic.

The convergence of four independent null spatial findings—centrality, POI density, activity zone comparison, and three-tier capacity analysis—provides robust evidence. This is consistent with induced demand theory (Duranton and Turner, 2011), though our study does not directly test causal mechanisms.

Normalization and absolute metrics. Because the jam factor normalizes speed to free-flow conditions, one might suspect the null spatial findings are a methodological artifact: if the metric already divides out capacity differences, then capacity cannot predict it. To address this, we re-aggregated the raw data with absolute speed and free-flow columns, computing two denormalized metrics: (1) raw speed (km/h), and (2) speed reduction ($v_f - v$), which measures excess travel time independent of segment-specific normalization. The temporal dominance pattern persists across all three metrics (Table 9): jam factor ($\eta^2 = 8.8$–9.4%), speed reduction ($\eta^2 = 8.8$–9.6%), and absolute speed ($\eta^2 = 5.1$–5.9%) all show strong temporal effects, while spatial predictors remain negligible. The weaker effect for raw speed reflects the limited range of speed variation when aggregated across diverse road types (motorways at 60 km/h versus residential streets at 20 km/h obscure temporal patterns), whereas speed reduction isolates the delay component. These results confirm that temporal dominance is not an artifact of normalization but a genuine characteristic of congestion dynamics.

6.2. Failure of spatial predictors

The near-zero correlation between spatial characteristics and congestion contradicts common planning assumptions. Neither topologically important roads (high betweenness) nor activity-dense areas (high POI density) predict congestion. This extends findings by Kirkley et al. (2018) questioning the centrality–congestion link and adds POI density and capacity effects to the null result.

The bottleneck analysis provides particularly counterintuitive evidence: three independent tests yield consistently null results ($p = 0.06$–0.83, $|d| < 0.09$). The jam factor normalization explains part of this—each segment is measured relative to its own capacity—but the spatial capacity drop test examines *relative* disruptions that should be detectable even under normalization if bottleneck queuing is a dominant mechanism.

6.3. Implications for traffic management

These findings suggest important policy directions, noting the caveat that they characterize relative congestion:

1. **Demand management over infrastructure expansion:** Staggered work/school hours, flexible work policies, and congestion pricing directly address temporal synchronization and could reduce peak congestion by 30–40%.
2. **Temporal targeting:** Concentrating traffic management resources (signal optimization, police deployment, incident response) during 16:00–19:00 maximizes impact.
3. **Infrastructure investment:** High-betweenness roads are not necessarily expansion priorities; bottleneck removal shows no measurable benefit in normalized congestion. Absolute delay analysis is needed before informing major infrastructure decisions.

6.4. Pipeline contribution and reproducibility

The pipeline's city-agnostic and provider-agnostic design—demonstrated across a 20× population range—enables researchers in data-scarce contexts to replicate sophisticated traffic analyses with minimal setup. A key architectural decision is the implementation of data collection as a pluggable provider system: an abstract `TrafficProvider` base class defines a common interface, and concrete implementations for HERE, TomTom, and Google handle API-specific details (authentication, query strategy, response parsing). This design

ensures that (1) the pipeline is not locked into a single commercial vendor, (2) researchers can choose whichever provider has better coverage or pricing in their region, and (3) adding a new provider requires implementing only the `fetch_flow()` method without modifying any downstream analysis code. The entire pipeline—from data collection to publication-ready figures—is implemented in pure Python, eliminating the need for R or other language runtimes. Key design features include: modular stages with well-defined inputs/outputs, standardized GeoPackage format, centralized configuration, and installation via PyPI. The pipeline has been validated across three cities with distinct morphologies (coastal megacity, highland basin, coastal–hillside hybrid), suggesting robustness to varying urban characteristics.

6.5. Limitations

1. **Metric sensitivity:** While we tested temporal dominance across normalized jam factor, denormalized speed reduction, and absolute speed (Table 9), each metric captures different aspects of congestion. Jam factor emphasizes relative delay, speed reduction measures absolute excess travel time, and raw speed reflects operational conditions. The consistent temporal dominance across all three strengthens the finding, though further investigation with travel time or queue length could reveal additional nuances.
2. **API coverage:** Commercial traffic APIs may underrepresent minor roads and informal settlements. While the multi-provider architecture allows cross-validation between providers, the present study uses HERE data exclusively. Provider-specific biases in road coverage and probe vehicle density could influence results and warrant further investigation.
3. **Capacity scoring:** Ordinal scores from OSM highway classification proxy actual capacity; lane count, signals, and intersection design are not captured.
4. **Segment length heterogeneity (MAUP):** Segments vary from $\sim$200m to $>$3km; length-weighted statistics could be explored (Openshaw, 1984).
5. **Temporal scope:** The 11-month period may not capture long-term trends.
6. **Causality:** Correlational findings do not establish causal relationships.

7. Conclusions

This study presents an open-source, pure-Python pipeline for spatiotemporal traffic congestion analysis integrating commercial traffic flow APIs (HERE, TomTom, Google) through a pluggable provider architecture, OSMnx, and PySAL. The pipeline is demonstrated across three Indonesian cities using 264 million observations over 11 months.

The primary empirical finding is that **relative congestion is fundamentally temporal**: time-of-day explains 15–24% of jam factor variance, while spatial predictors (centrality, POI density, road capacity, capacity drops) explain <0.02% combined—a 1,000–4,000× gap. Four independent spatial tests yield null results, demonstrating that congestion patterns reflect synchronized demand, not problematic locations.

The pipeline is available as `pip install traffic-congestion-pipeline` (https://pypi.org/project/traffic-congestion-pipeline/) with source code, API documentation (https://firmanhadi21.github.io/traffic-analyses/), and aggregated datasets on Zenodo (DOI: 10.5281/zenodo.18650759). It requires only a bounding box and API key to replicate for any city worldwide.

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Declaration of competing interests

The authors declare no competing financial interests or personal relationships that could have influenced this work.

CRediT authorship contribution statement

Firman Hadi: Conceptualization, Methodology, Software, Data curation, Formal analysis, Writing – original draft. **Yasser Wahyuddin:** Methodology, Formal analysis, Writing – original draft. **L.M. Sabri:** Formal analysis, Writing – review & editing. **Agung Indrajit:** Writing – review & editing.

Declaration of generative AI

During the preparation of this work, the authors used Claude (Anthropic) for data processing pipeline development, evaluation script implementation, and statistical analysis automation. The authors reviewed and edited all outputs and take full responsibility for the content of this publication.

Data and software availability

Software. The pipeline is released as an installable pure-Python package: `pip install traffic-congestion-pipeline` (https://pypi.org/project/traffic-congestion-pipeline It supports HERE, TomTom, and Google traffic APIs through a pluggable provider interface. Source code is available under the MIT licence at https://github.com/firmanhadi21/traffic-analyses. API documentation: https://firmanhadi21.github.io/traffic-analyses/.

Dataset. Aggregated traffic data (24 GeoPackage files, ∼115 MB) are archived on Zenodo: https://doi.org/10.5281/zenodo.18650759. Researchers can reproduce all figures with:

```
pip install "traffic-congestion-pipeline[all]"
traffic-pipeline geostatistics
traffic-pipeline bottleneck
traffic-pipeline poi
traffic-pipeline synthesis
```

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Supplementary material

Supplementary tables S1–S5 (spatial weight sensitivity analysis, period-specific Moran's I, Getis-Ord Gi* results, spatial match quality, spatial regression results), additional figures (hotspot maps for Bandung and Semarang, capacity scatter plots, street orientation polar plots), and full statistical outputs are available in the online supplementary material and at https://github.com/firmanhadi21/traffic-analyses.